# CSC8370 Data Security

## 1. Project Overview

Students are required to choose one of three levels of project difficulty for MNIST Classification. Need to complete the selected difficulty level and previous level(s), for example if choose level 3, levels 1 and 2 also need to complete. Each level builds upon the previous, introducing more complexity and challenges related to security/privacy in machine learning. You can use the provided servers for training and testing the models. The three levels are described below:

### Level 1: Centralized Learning for MNIST Classification

In this level, students will implement a centralized machine learning system for classifying handwritten digits from the MNIST dataset.

- Task: Implement a deep learning model that classifies **MNIST** images.
- Requirements:
  - o Use a neural network model to classify the images.
  - o Train the model in a centralized manner (i.e., all data is in one location).
  - o Achieve a reasonable level of accuracy for the classification task.

### Level 2: Federated Learning for MNIST Classification

This level requires students to implement federated learning with 10 clients, each holding a portion of the MNIST dataset.

- Task: Implement a federated learning version.
- Requirements:
  - o Implement federated learning with **10 clients**.
  - o Each client should independently train on its own local data (a partition of the MNIST dataset).
  - o Clients should share model parameter updates, which are aggregated in a central server.
  - o Ensure the model achieves a reasonable level of accuracy after several rounds of training.

### Level 3: Robust Federated Learning with Attack Detection

This advanced level introduces a malicious client that uploads incorrect model updates to disrupt the federated learning process. *The goal* is to detect this client and minimize the impact of their false updates.

- Task: Implement robust federated learning with attack detection.
- Requirements:
  - o Simulate a scenario where one of the 10 clients becomes malicious from a certain round onward.
  - o The malicious client should submit falsified model updates (e.g., random or incorrect parameters).
  - o Implement a mechanism to detect the malicious client.
  - o Minimize the impact of the malicious client on the overall model.
  - o Ensure the learning process remains robust and accurate despite the presence of an attacker.
  - o It is stipulated that the training phase only run 10 epochs.

## 2. Project Requirements

The requirements of the final project are listed as follows:

- Project Proposal

Every group contains *1-2 students* and selects a project level and *report the project proposal (1-2 pages), including the project level, participant(s), potential solutions in your selected project level, experiment design*, to iCollege by *Nov. 04, 2025*.The project level could be changed before demo day.

- Source Code:
    - Should be submitted to the TA by *Nov 25, 2025*.
    - Include a clear **README.md** file with instructions on running the code and the required environment versions. It is recommended to use PyTorch for designing the code.
    - The code should include functionality for *saving model parameters locally and loading them* to resume training or perform model testing. (Guide: link)
    - Ensure that *the best-performing model parameters are saved*, which can be used by the TA to test the model's performance.

- Final Project Report: Should be submitted by *Nov 25, 2025*, ***with no late submissions accepted.***

## 3. Project Grading Policy

The project grade consists of three components:

- Level Selected:
    - Maximum 10 points for completing Level 1.
    - Maximum 18 points for completing Level 2.
    - Maximum 30 points for completing Level 3.

- Project Report (40% points):

  The report will be graded over 100 points. It should include the following sections (5-10 pages):
    - (1) Introduction (10 points): background and motivation, problem definition, system overview, etc.
    - (2) Methodology (40 points): algorithms. Clearly describe the steps of manually implementing framework.
    - (3) Evaluation (30 points): settings for experiments and demo, observations and inferences.
    - (4) Learning outcomes (20 points): tasks accomplished by each team member, reasons for unaccomplished tasks or any failures, lessons learned from the project and teamwork experience.

- Classification accuracy ranking (60% points):
    - During Demo Day, levels 1 and 2 will be tested ***using our provided dataset*** (All datasets are IID).
    - For levels 1 and 2, we only run the test phase, which loads the saved model parameters (trained by the students themselves) and uses the dataset we provide (independent and identically distributed with the MNIST dataset) to see the final accuracy.
    - First, the code will be tested to ensure it runs correctly.
    - Afterward, ***the pre-trained model parameters*** will be loaded, and model performance will be evaluated using the provided dataset.
    - Scores will be based on the model's performance (**accuracy**) and ranked from highest to lowest.
    - For level 3, we will run the entire training phase for 10 epochs to check whether malicious client detection has been implemented and to see the final accuracy.

**4. Project Code and Model Sharing Policy**

It is ***strictly prohibited*** for different groups to share project models or code, if you want to select **the level 3**. Any resemblance in code or model parameters, especially if the size of the saved model parameters is identical, will be flagged during Demo Day.

- Explanation Requirement

If such similarities are detected, the group(s) involved must provide a thorough explanation. During the presentation, you will be required to explain the specific functionalities of each model component and demonstrate the originality of your implementation.

**5. Dataset Instructions**

- Level 1 (Centralized Learning)

If you are implementing Level 1, please MUST use this dataloader to load the dataset. The link provides instruction on how to integrate the dataloader into your project, especially if you are using the PyTorch framework.

- Level 2 & Level 3

For those implementing Level 2 or Level 3, please MUST refer this dataloader to load the dataset. This link also includes a guide on how to correctly use the dataloader in a federated setting.

**6. Project submission requirements and deadlines**

| Deliverable/Submission | Deadline | Assessment |
|---|---|---|
| Proposal –submit to assignment-Project proposal | 23:59PM, Nov. 04, (Tue) | - |
| (1) Source Code Package<br>(2) Final Project Report – submit to assignment-Project Code package and report | 23:59PM, Nov. 25, (Tue) | 40% |
| Demo Day – In classroom | 2:45PM, Dec 2, (Tue) | 60% |